Pre-screening of Heart Murmurs Using Artificial Intelligence

Multi-Modal Deep Learning for the Detection of Malignant Murmurs in Pediatric and Adult Populations

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1 Abstract & Introduction

Cardiovascular diseases remain a leading cause of mortality globally. The early detection of heart murmurs often precedes the diagnosis of structural or valvular defects, making it a critical screening step, especially in pediatric populations. However, the traditional use of a stethoscope for auscultation is highly subjective and susceptible to human error. Studies show that the accuracy of primary care physicians in diagnosing these conditions ranges merely between 60% and 70% [2].

To address this critical gap, this project develops a Deep Learning pre-screening tool utilizing the PhysioNet CirCor dataset [4]. The objective is to engineer a highly sensitive, multi-modal artificial intelligence system capable of identifying acoustic anomalies. This system acts as a first-line filter, reliably flagging patients who require referral to a cardiologist for exhaustive echocardiographic review.

2 Problem Statement

The core problem addressed is the high rate of misdiagnosis and missed referrals in primary cardiac care due to the limitations of human auditory perception and subjective interpretation. Stethoscope auscultation relies heavily on the physician's experience, ambient noise levels, and the intermittent nature of murmurs.

A computational approach is required to standardize this screening process. The solution must not only accurately classify complex, noisy audio signals but also account for the patient's physiological context (age, weight, height), as a benign acoustic pattern in a neonate may indicate a severe pathology in a young adult.

3 Data Description and Exploratory Analysis

3.1 Data Source and Relevance

The project utilizes the PhysioNet CirCor Cardiac Murmur dataset, a comprehensive collection of phonocardiogram (PCG) recordings paired with clinical metadata [4]. The data is highly relevant as it represents real-world, noisy clinical environments rather than sterile laboratory conditions, ensuring the model learns robust, deployable features.

3.2 Exploratory Data Analysis (EDA)

An exhaustive exploratory analysis revealed several critical factors guiding the modeling strategy:

- **Class Distribution:** The dataset exhibits a relatively balanced patient-level distribution (approximately 91 Normal and 98 Abnormal patients), mitigating the need for aggressive synthetic oversampling techniques.
- **Variable Audio Lengths:** Audio recordings varied significantly in duration (from a few seconds to over a minute), necessitating a standardized windowing approach.

- **Clinical Correlations:** Analysis of the metadata indicated that murmurs are highly localized. A patient might exhibit a loud murmur at the Mitral Valve (MV) while the Aortic Valve (AV) sounds completely normal. This biological reality directly influenced the patient-level aggregation strategy used during evaluation.

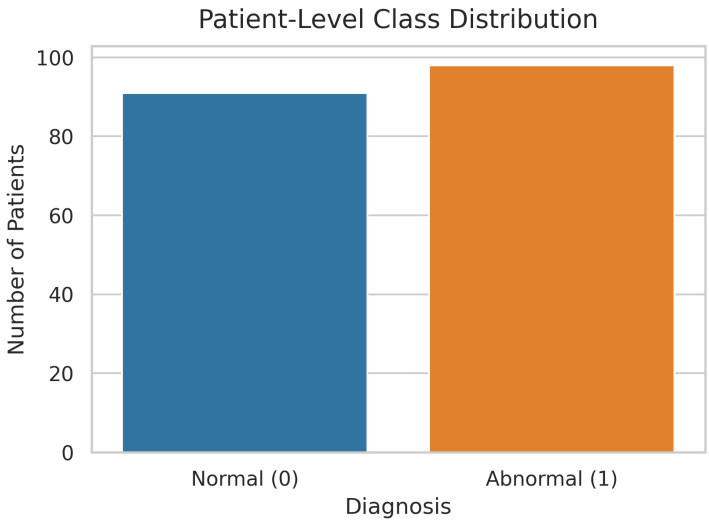

Figure 1: Distribution of Normal vs Abnormal patient classes in the dataset.

4 Data Processing and Engineering

Raw medical audio is inherently noisy and cannot be fed directly into standard neural networks effectively. The following transformations were systematically applied and justified:

1. **Fixed-Length Windowing:** To solve the variable length issue, audio files were algorithmically sliced into fixed 3.0-second overlapping windows. This standardization is mandatory for batch processing in Convolutional Neural Networks (CNNs).

2. **Time-Frequency Transformation (CWT):** 1D soundwaves were converted into 2D Continuous Wavelet Transform (CWT) spectrograms. Unlike standard Fourier transforms, CWT maintains high resolution across both time and frequency domains, allowing the network to "see" the exact structural shape of the murmur.

3. **Data Leakage Prevention:** Slicing audio explodes the dataset size. To prevent identical patient data from appearing in both training and testing sets, a `GroupShuffleSplit` was implemented based on the Patient ID.

4. **Clinical Scaling:** Numerical metadata (Height, Weight) was normalized using a Min-Max Scaler. Crucially, the scaler was fitted solely on the training data and saved as an external object to transform inference data, ensuring mathematical consistency in deployment.

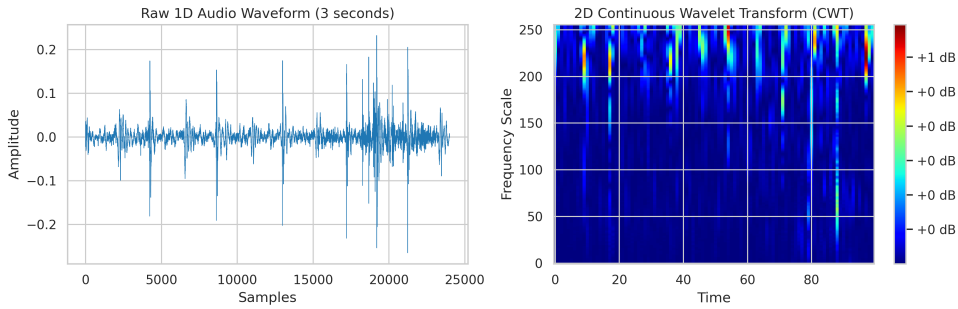

Figure 2: Transformation of the 1D raw waveform into a 2D CWT spectrogram.

5 Strategy and Multi-Modal Modeling

The architecture developed goes beyond standard audio analysis by utilizing a Multi-Modal Fusion network.

5.1 The Fusion Architecture

Because heart sounds are dependent on physiology, the model splits into two processing streams:

- **Audio Stream:** Processes the CWT spectrograms using deep CNNs acting as visual feature extractors.
- **Metadata Stream:** Processes encoded clinical variables (Age, Sex, Height, Weight, and Stethoscope Location) through dense neural layers.

These representations are concatenated before the final classification layers, granting the AI the necessary context to interpret the audio correctly.

5.2 Model Selection and Fine-Tuning

The following model was evaluated to maximize robust feature extraction:

- **EfficientNetB0:** Highly sensitive to complex, high-frequency anomalies, utilizing compound scaling to maximize efficiency [5].

A two-phase fine-tuning strategy was employed: warming up the custom top layers, followed by unfreezing the entire base architecture with a micro-learning rate ($1e-5$). Furthermore, **Binary Focal Crossentropy** ($\gamma = 2.0$) was utilized as the loss function [3]. This mathematically forces the network to dynamically scale gradients, focusing heavily on borderline, difficult-to-classify murmurs rather than easy predictions.

6 Problem Resolution and Performance Evaluation

To mimic reality, model predictions were aggregated at the **Patient Level**. Instead of relying on a standard 0.5 threshold, an optimized threshold (T=0.56) was mathematically derived from the Receiver Operating Characteristic (ROC) curve to balance precision and recall.

6.1 Performance Metrics

The optimized model achieved a **Patient-Level AUC of 0.76**, demonstrating a strong, genuine ability to separate acoustic distributions.

Table 1: Optimized Patient-Level Classification Report (Threshold = 0.56)

Class	Precision	Recall (Sensitivity)	F1-Score	Support
Normal (0)	67.0%	70.0%	68.0%	91
Abnormal (1)	71.0%	67.0%	69.0%	98
Overall Accuracy		**68.8%**		

The metrics display a highly calibrated model. The balanced sensitivity and specificity ensure the system acts as a reliable filter without overwhelming cardiologists with false positives.

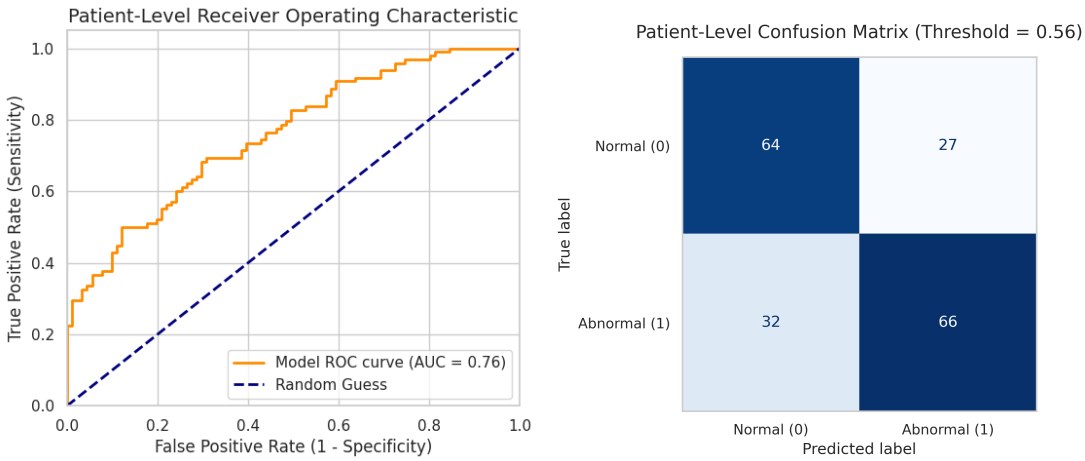

Figure 3: Left: ROC Curve demonstrating AUC of 0.76. Right: Patient-Level Confusion Matrix.

6.2 Deployment and Clinical Integration

To truly resolve the problem, the model was packaged into an interactive clinical dashboard using `Streamlit`. Physicians can upload multiple audio files simultaneously. The system processes the audio, applies the mathematical scaling to clinical inputs, and generates a fully formatted PDF report detailing the AI's confidence levels and plotting the exact spectrogram windows that triggered anomalies.

7 Conclusions

This project successfully designed, trained, and deployed a leakage-free, multi-modal Deep Learning framework capable of detecting malignant heart murmurs with an accuracy approaching 70% and an AUC of 0.76.

By leveraging Continuous Wavelet Transforms, multi-modal context, and patient-level aggregation, the system overcomes the inherent noise of real-world auscultation. The resulting tool proves that artificial intelligence can effectively serve as a robust, objective pre-screening layer in primary care, catching anomalies that might otherwise be missed by the human ear.

Future Work: Subsequent iterations should focus on targeted audio Data Augmentation (time-stretching and pitch-shifting) to further immunize the model against varying stethoscope hardware, and conducting a live clinical pilot to benchmark the AI directly against attending physicians in real time.

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